

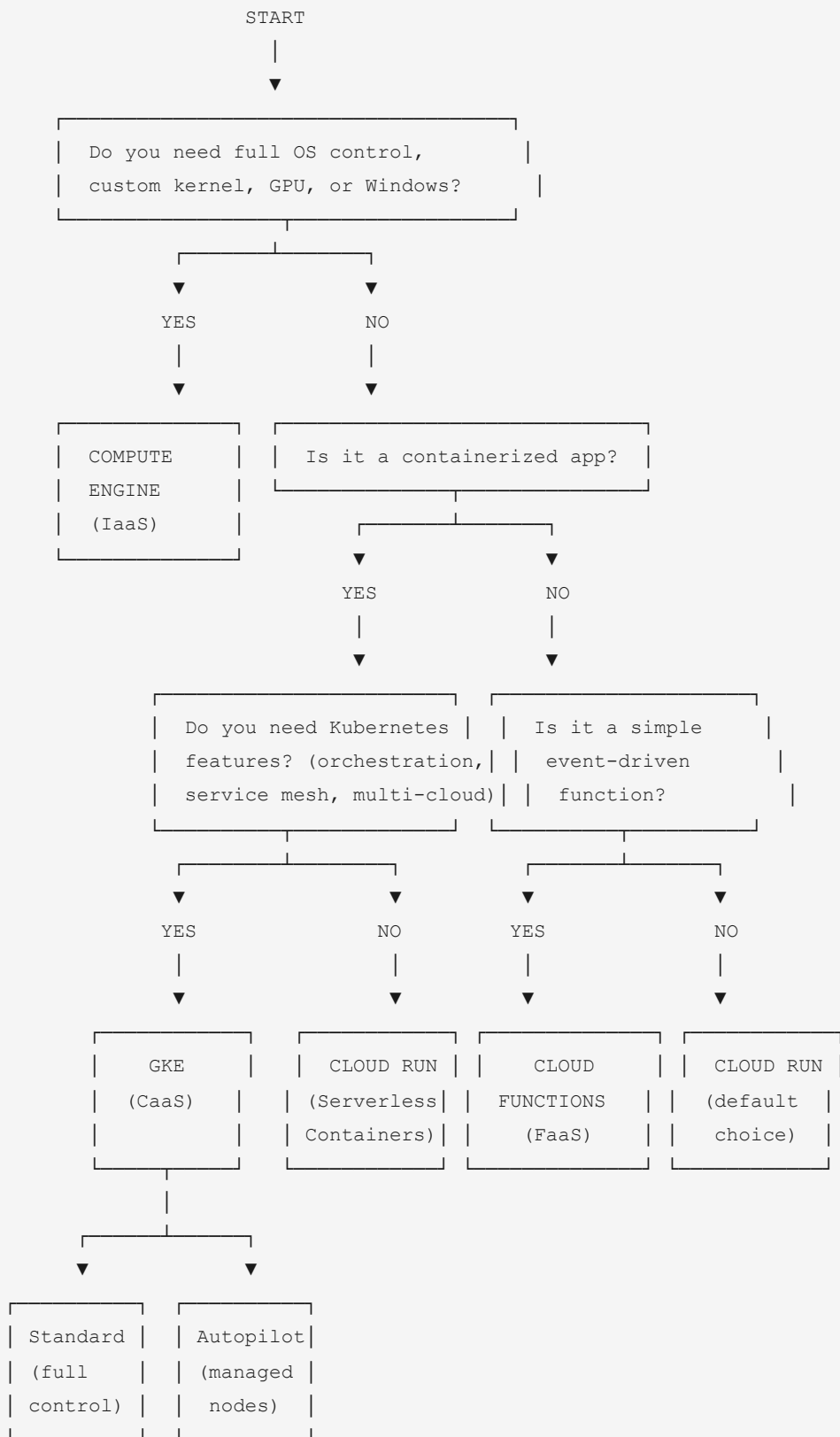
GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 2 -- Visual Maps: Planning and Configuring a Cloud Solution

GCP Study Notes

Study Notes April 2026

1. Compute Decision Tree



2. Compute Options – Comparison Matrix

	Compute Engine	GKE	Cloud Run	Cloud Functions
Model	IaaS (VMs)	CaaS (Containers)	Serverless (Containers)	FaaS (Functions)
You manage	OS, runtime, app, config	Containers, K8s config	Container image only	Function code only
Scaling	Manual or autoscaler	Pod + Node autoscaler	Auto (0 to N)	Auto (0 to N)
Startup time	Minutes	Seconds	Seconds	Seconds
State	Stateful	Both	Stateless	Stateless
Max timeout	Unlimited	Unlimited	60 min	60 min (gen2)
Pricing	Per second (min 1 min)	Cluster fee + node VMs	Per request + CPU/mem	Per invocation + compute
Scale to zero	No	No	YES	YES
Best for	Legacy, GPU, Windows, lift-shift	Microservices hybrid/multi- cloud, K8s	Stateless web apps, APIs	Event-driven webhooks, triggers

Control vs Management Spectrum

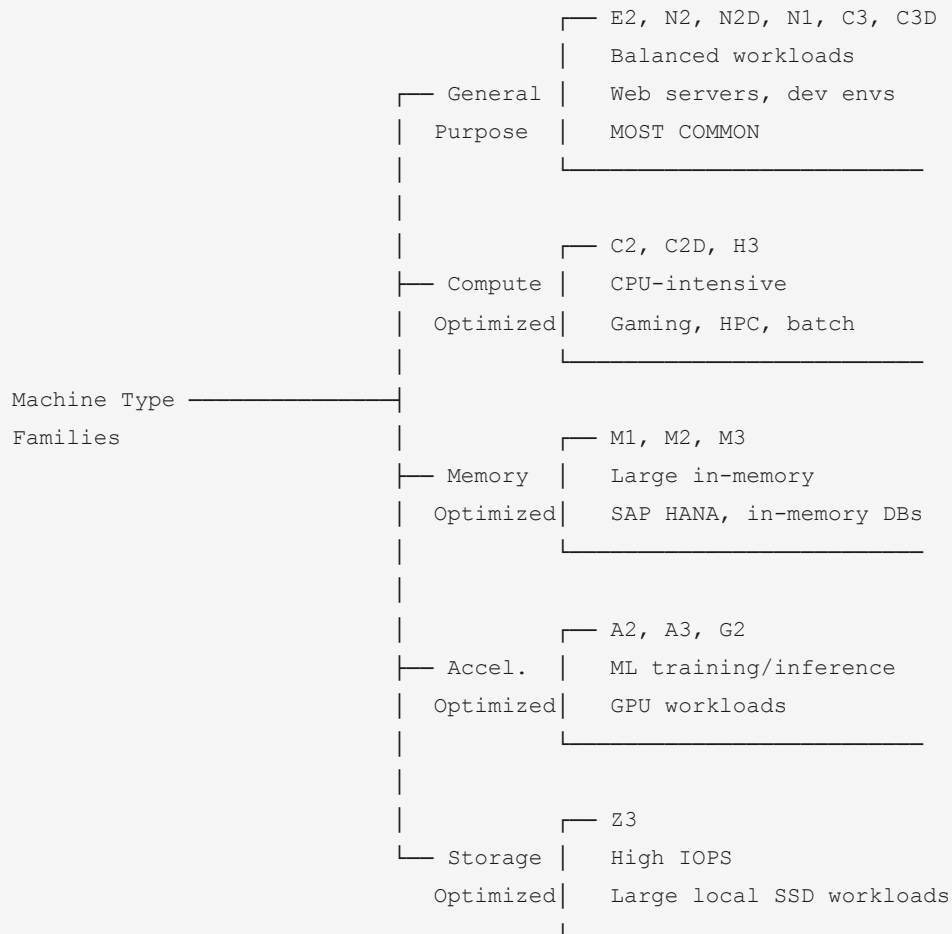
MORE CONTROL
(you manage)

MORE MANAGED
(Google manages)



Compute Engine	GKE	Cloud Run	Cloud Functions
OS	Containers	Container	Function
Runtime	K8s config	image	code
App	Node pools	(that's it)	(that's it)
Network			
Storage			
IaaS	CaaS	Serverless	FaaS

3. Machine Type Families Mind Map

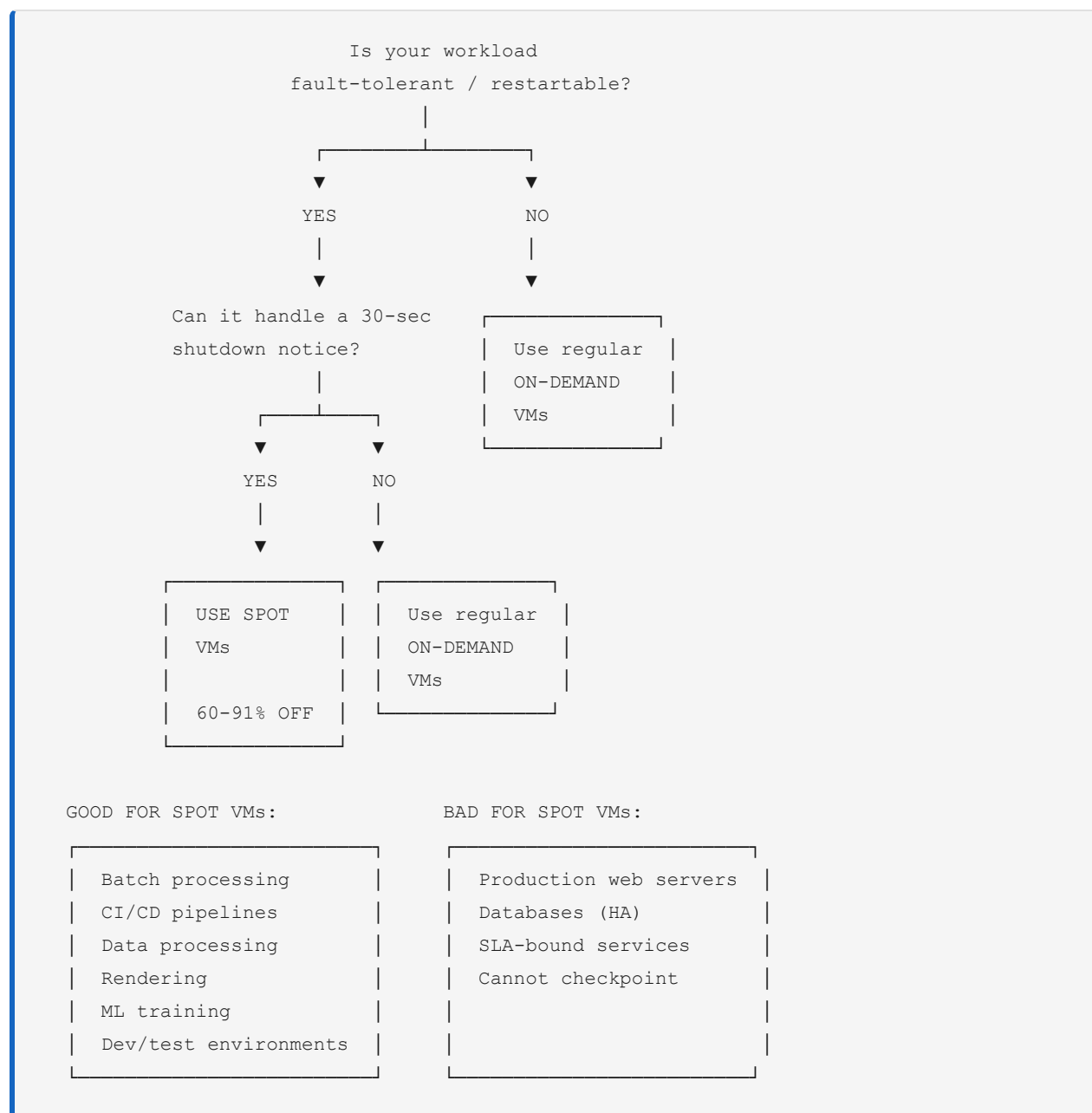


NAMING CONVENTION:

```

e2 - standard - 4
├── e2: Family prefix
├── standard: CPU:Memory ratio
│   ├── standard = 4 GB/vCPU
│   ├── highmem = 8 GB/vCPU
│   └── highcpu = 1 GB/vCPU
└── 4: 4 vCPUs
  
```

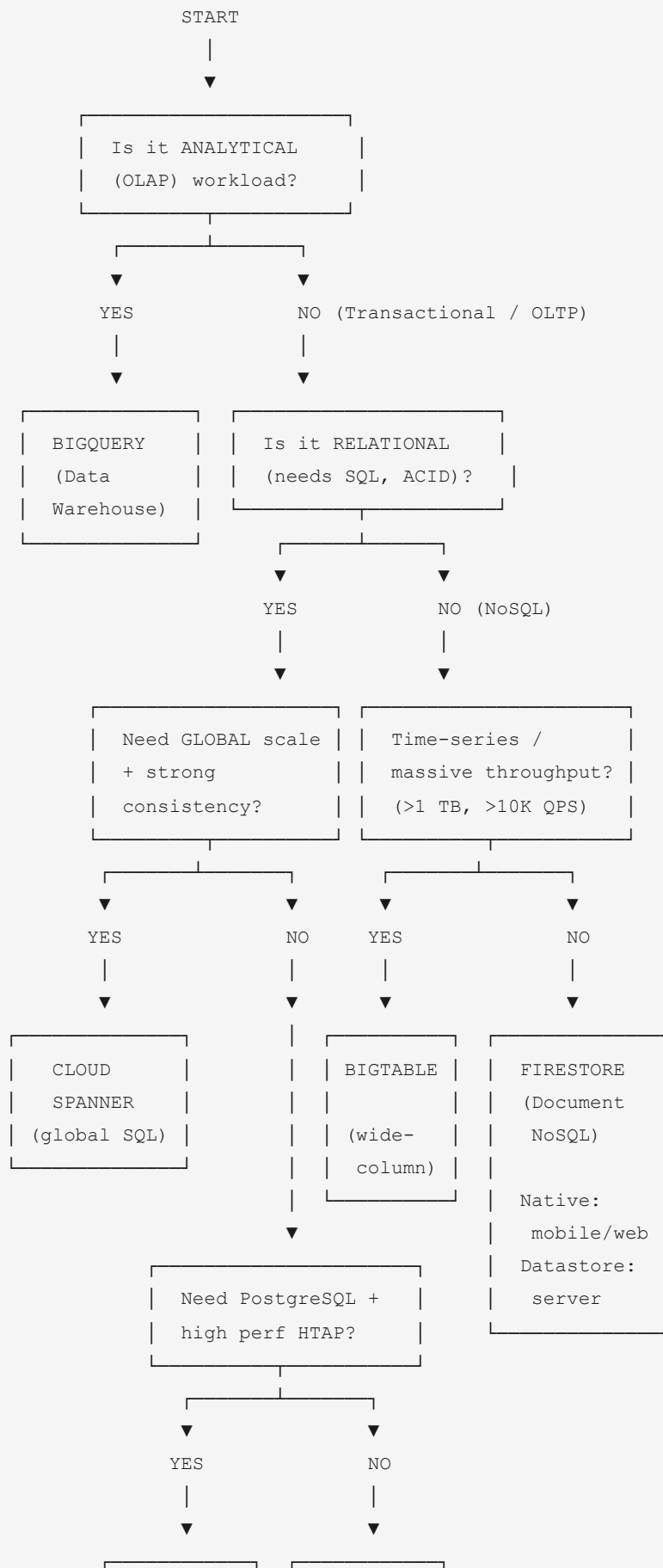
4. Spot VMs Decision Flowchart



The diagram illustrates the Compute Engine Pricing Ladder, ordered from MOST EXPENSIVE on the left to LEAST EXPENSIVE on the right.

On-Demand	SUDs (automatic)	CUDs (commitment)	Spot VMs (preemptible)
Full price	Up to 30% off	Up to 57% (1yr) or 70% (3yr) off	60-91% discount
Per second			Can be reclaimed
Min 1 min	25%+ usage per month		30-sec warning
No commitment Full flexibility	No commitment Automatic	1 or 3 year Manual purchase	No commitment No availability SLA

6. Database Decision Tree

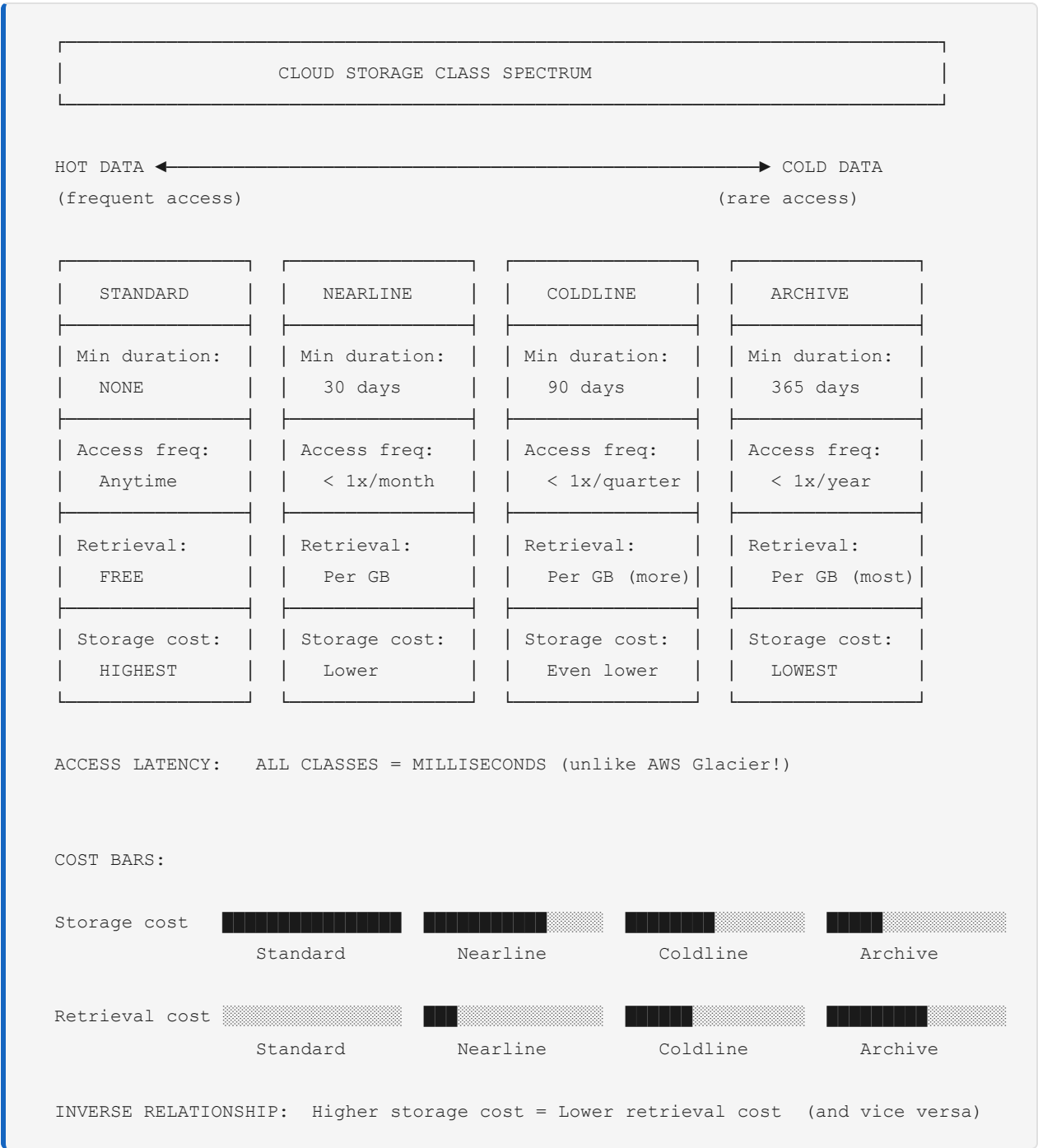


ALLOYDB	CLOUD SQL
(PG-compatible)	(MySQL, PG
HTAP)	SQL Srv)

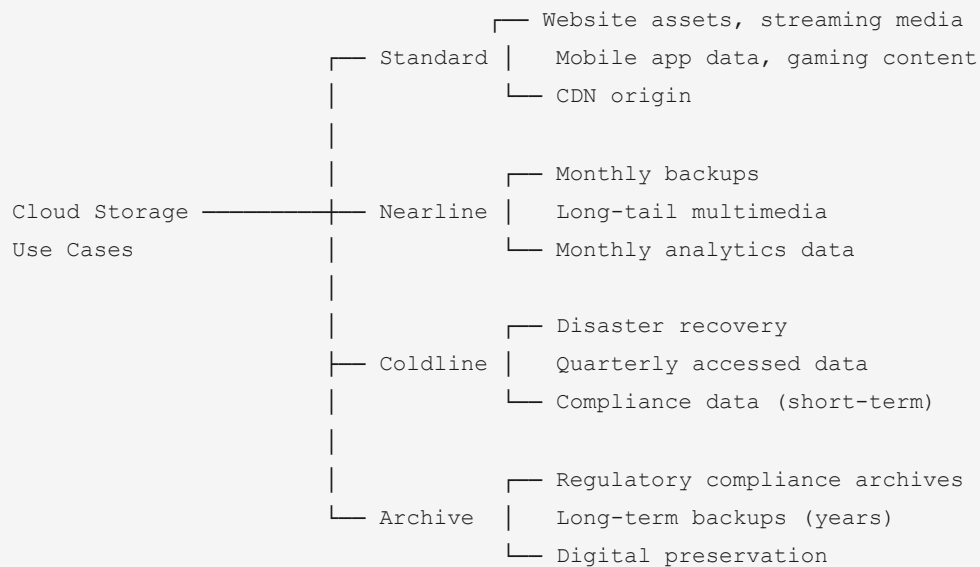
7. Database Comparison Matrix

	Cloud SQL	AlloyDB	Spanner	BigQuery	Firestore	Bigtable
Type	Relational	Relational	Relational	Analytical (DW)	Document (NoSQL)	Wide-column (NoSQL)
Workload	OLTP	OLTP+OLAP	OLTP	OLAP	Mobile/Web	Time-series
Scaling	Vertical	Vertical	Horizontal	Serverless	Serverless	Horizontal
Max Data	64 TB	128+ TB	Unlimited	Unlimited	1 TB/db	Unlimited
SQL	Full SQL	Full PG	Google SQL	Std SQL	No	No
Consistency	Strong	Strong	Strong (GLOBAL)	Eventual	Strong	Eventual
SLA	99.95- 99.99%	99.99%	99.999% (5 nines)	99.99%	99.999% (5 nines)	99.99%
Cost	Low	Medium	HIGH	Free tier	Free tier	Medium
Engines	MySQL PostgreSQL SQL Server	PostgreSQL	Proprietary	Proprietary	Proprietary	HBase A

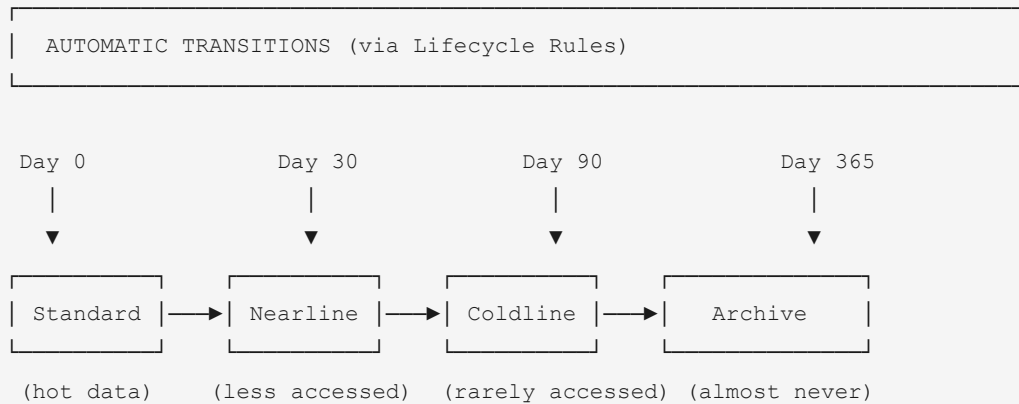
8. Cloud Storage Class Spectrum



Storage Class Use Cases



Object Lifecycle Management



Set lifecycle rules on the bucket to automate these transitions.
Can also auto-DELETE objects after a certain age.

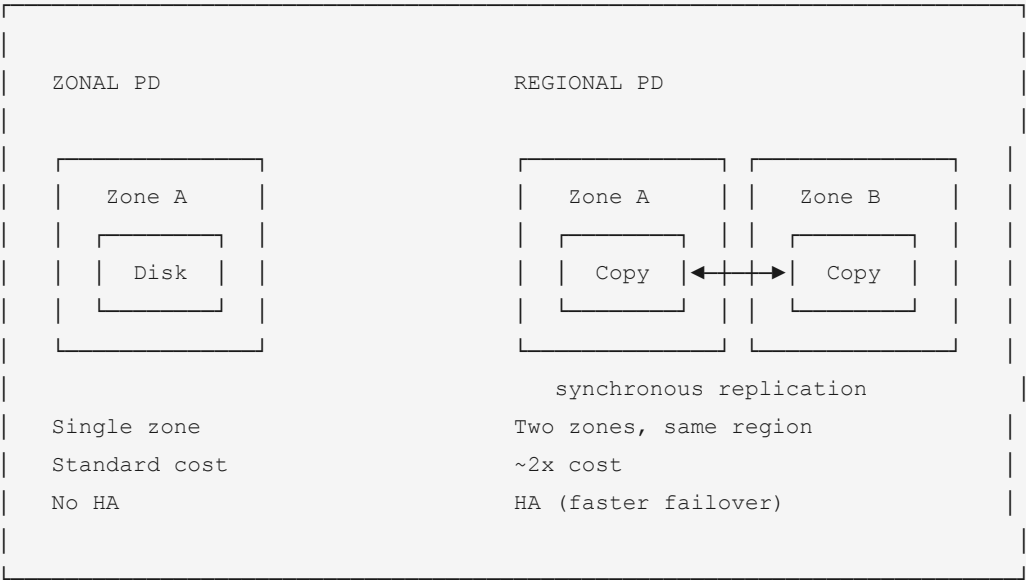
9. Block and File Storage Options

STORAGE TYPE OVERVIEW			
Object Storage (Cloud Storage)	Block Storage (Persistent Disk)	Block Storage (Local SSD)	File Storage (Filestore)
Unstructured data (files, images, backups)	Attached to VMs Durable Survives VM delete	Attached to host machine EPHEMERAL Data lost on VM stop/delete	NFS shared file system Multiple VMs can access

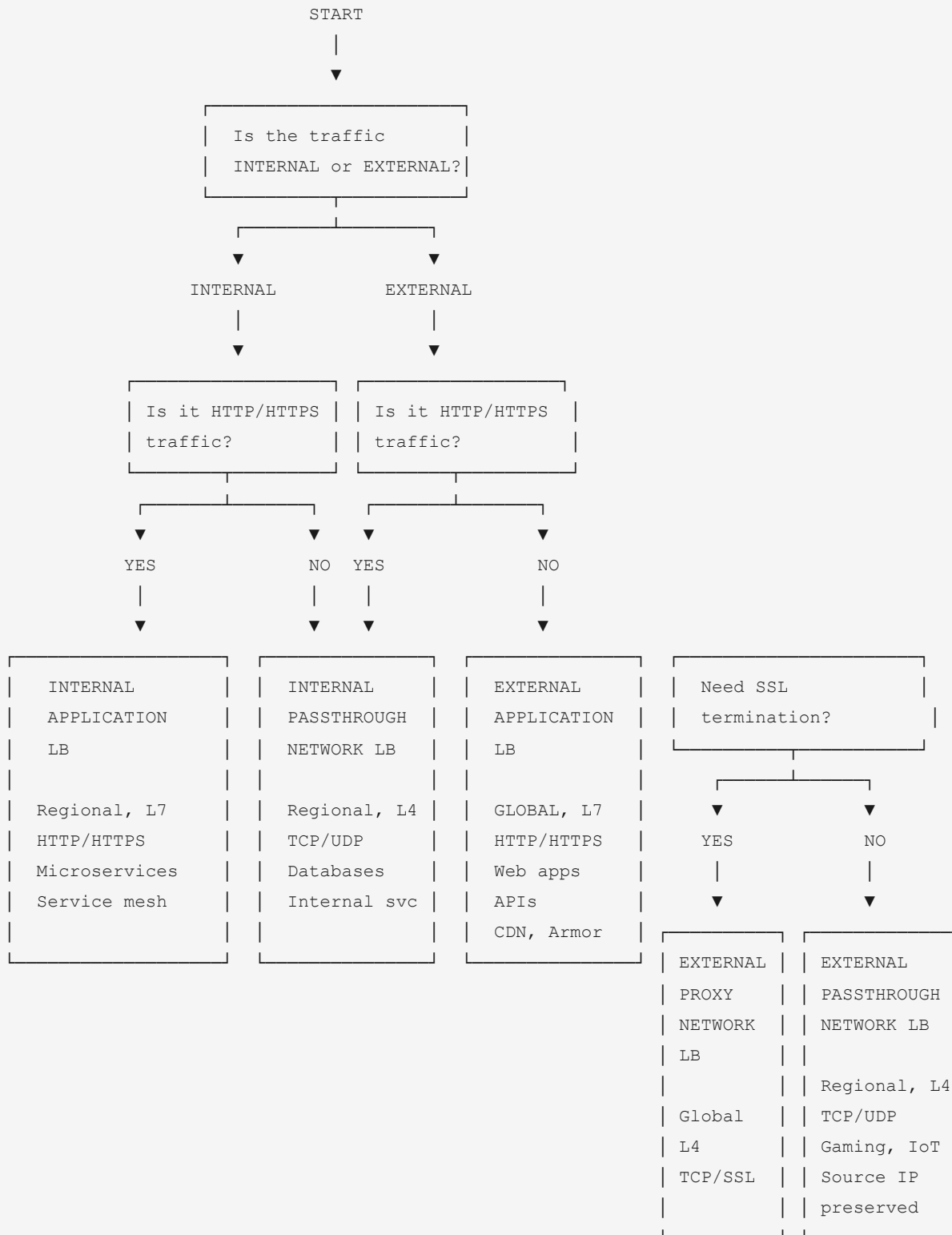
Persistent Disk Types

	pd-standard	pd-balanced	pd-ssd	pd-extreme
Type	HDD	SSD	SSD	SSD
IOPS	Low	Medium	High	Highest
Throughput	Low	Medium	High	Highest
Cost	Cheapest	Moderate	Higher	Highest
Use case	Bulk storage Sequential I/O	General purpose (DEFAULT)	Databases Random I/O	Mission- critical DBs

ZONAL vs REGIONAL PD:



10. Load Balancer Selection Flowchart



Load Balancer Comparison Matrix

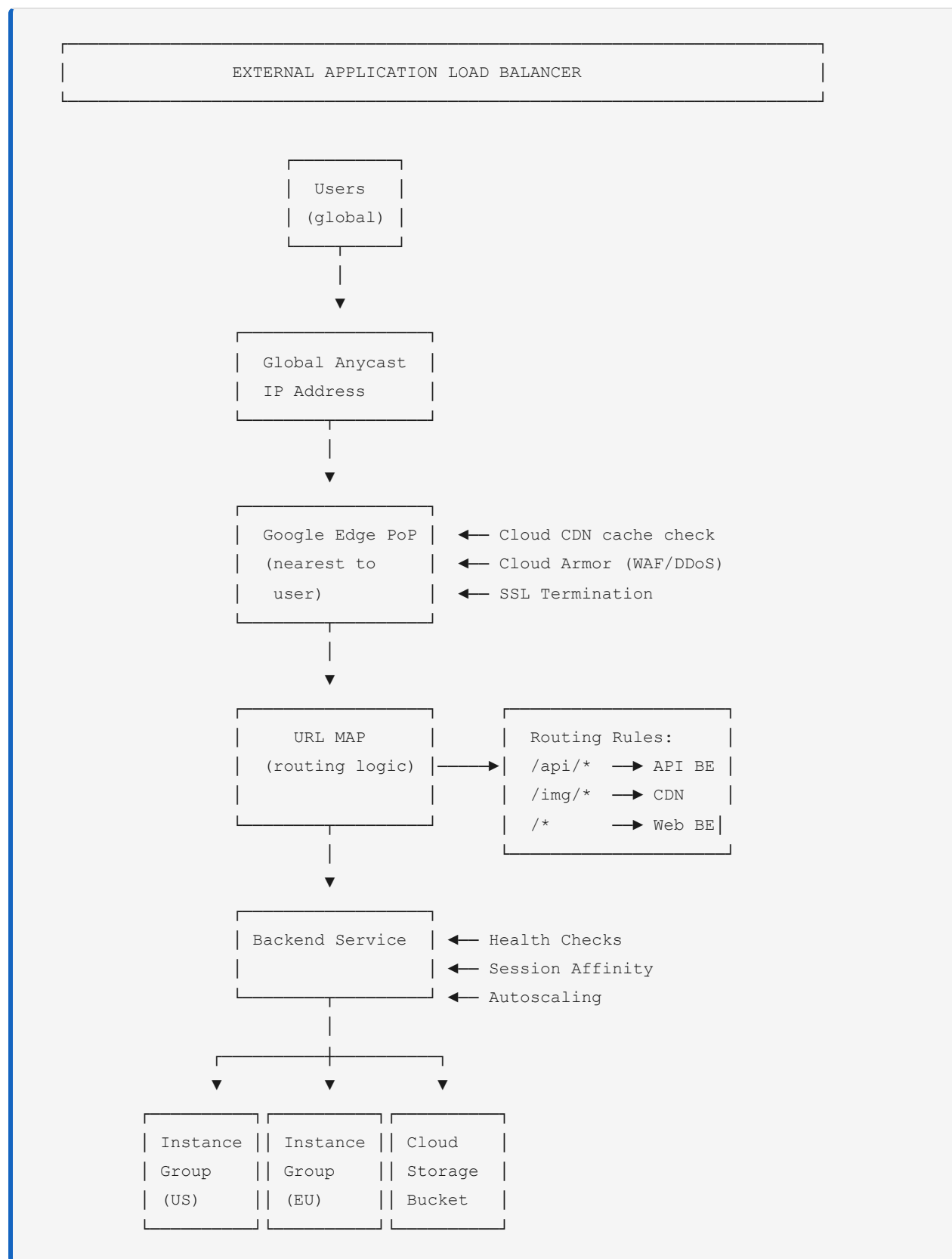
Load Balancer	Scope	Layer	Traffic	Key Use Case
Ext. Application LB	GLOBAL	L7	HTTP/S	Web apps, APIs, CDN, Armor
Int. Application LB	Regional	L7	HTTP/S	Internal microservices
Ext. Proxy Network	GLOBAL	L4	TCP/SSL	SSL termination, non-HTTP
Int. Proxy Network	Regional	L4	TCP	Internal TCP services
Ext. Passthrough Net	Regional	L4	TCP/UDP	Gaming, IoT, preserves IP
Int. Passthrough Net	Regional	L4	TCP/UDP	HA databases, internal svc

EXAM TIP: "Global web app" = External Application LB (almost always the answer)

EXAM TIP: "Internal microservices over HTTP" = Internal Application LB

EXAM TIP: "UDP traffic" = Passthrough Network LB (only one that supports UDP)

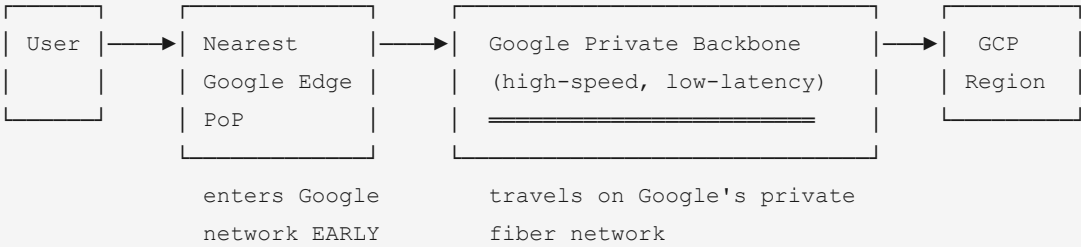
External Application LB Architecture



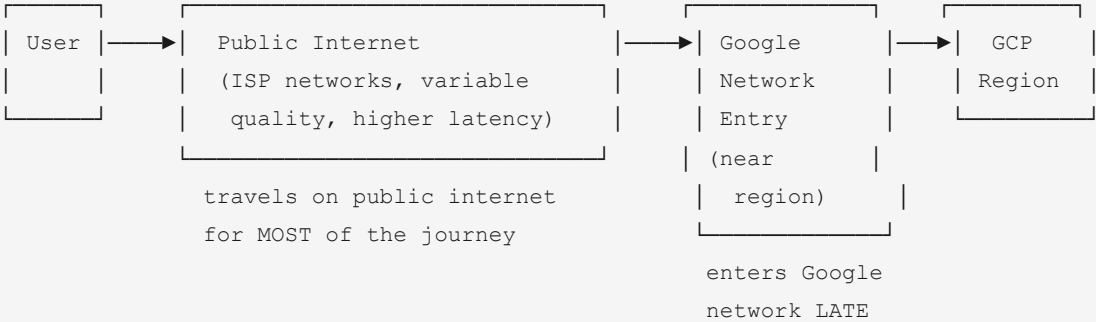
11. Network Tier Comparison

NETWORK SERVICE TIERS

PREMIUM TIER (Default)

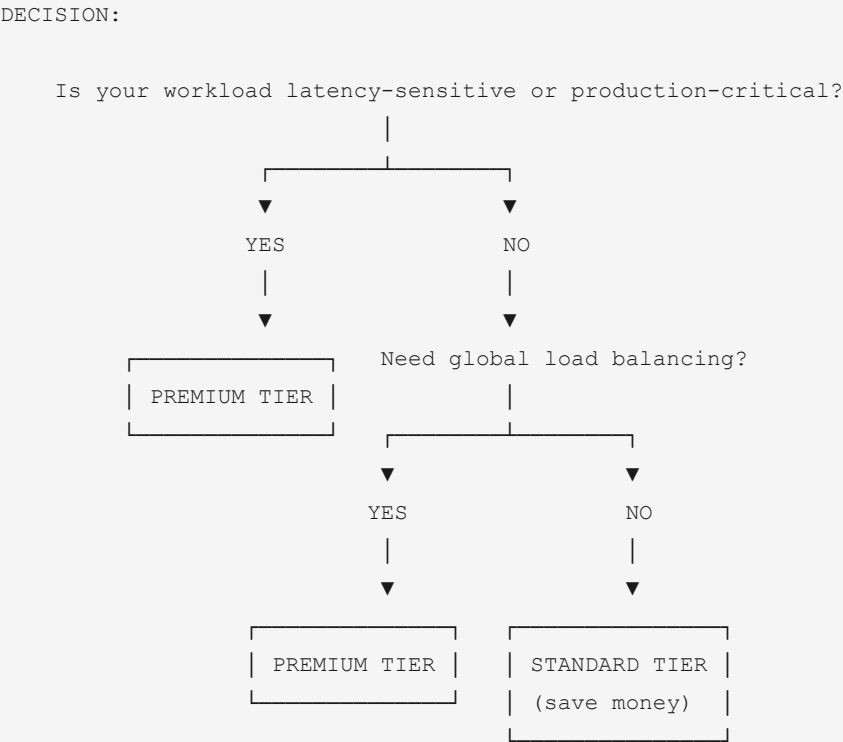


STANDARD TIER

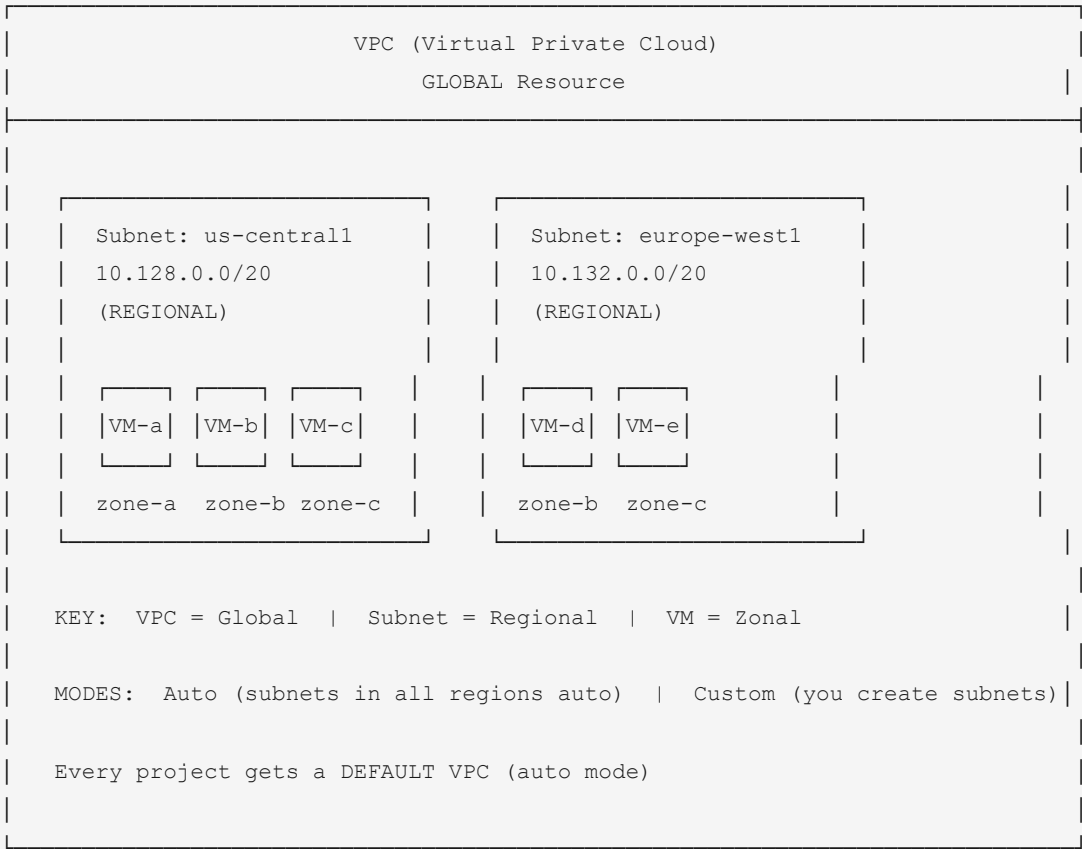


COMPARISON TABLE:

Feature	PREMIUM TIER	STANDARD TIER
Routing	Google backbone	Public internet
Latency	LOWER	Higher
Reliability	HIGHER	Variable (ISP)
Global LB	YES	NO (regional only)
SLA	Google network SLA	ISP-dependent
Cost	Higher	LOWER
Default	YES	No
Best for	Production	Dev/test
	Latency-sensitive	Cost-sensitive



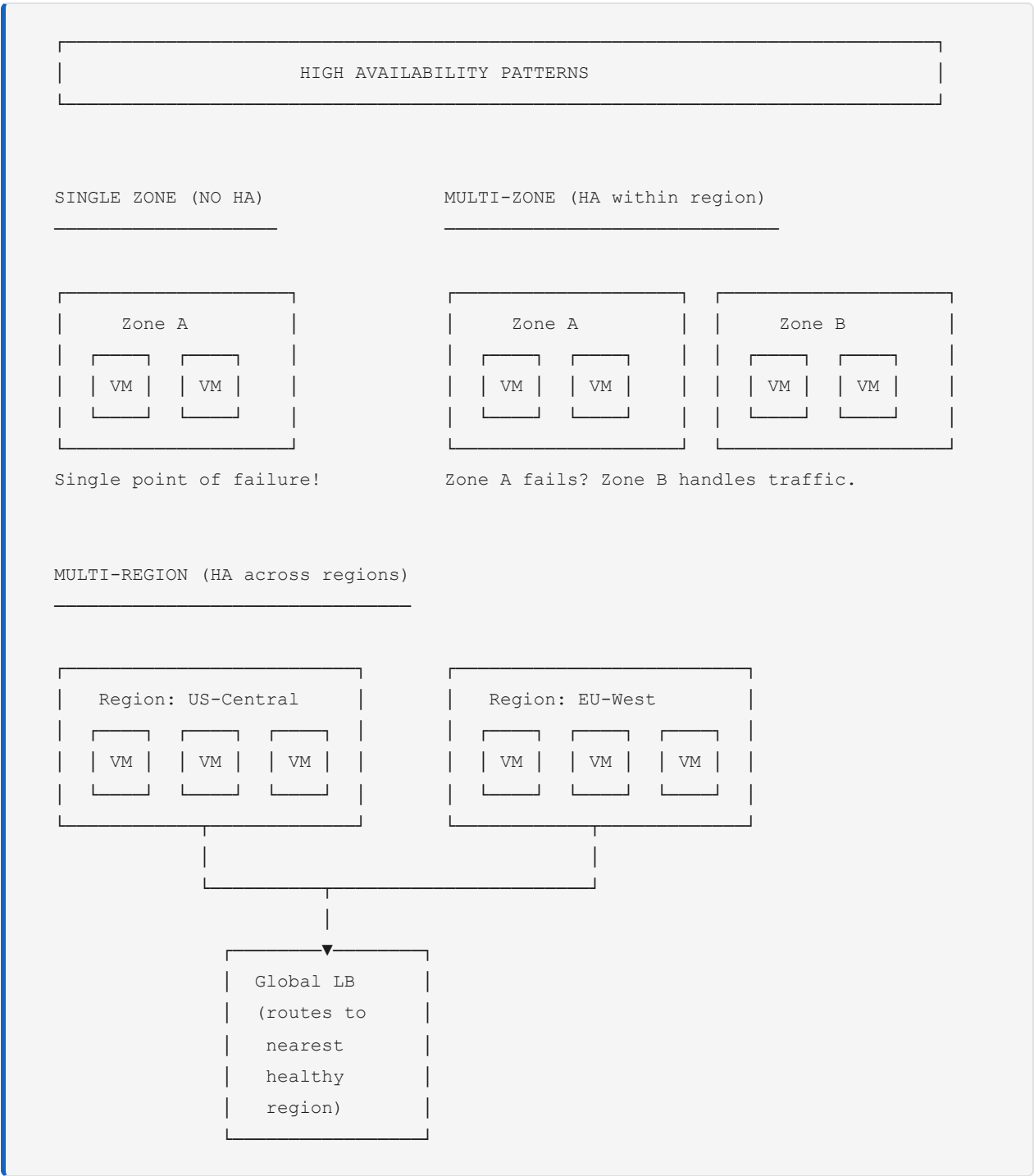
12. VPC Architecture



Resource Scope Reference

RESOURCE SCOPE	
GLOBAL	VPCs, Firewall rules, Routes, External IPs, Images, Snapshots, Global LBs, Cloud CDN
REGIONAL	Subnets, Regional PDs, Regional instance groups, Cloud Router, Cloud NAT, Internal IPs
ZONAL	VM instances, Zonal PDs, GKE node pools, Local SSDs

13. High Availability Patterns



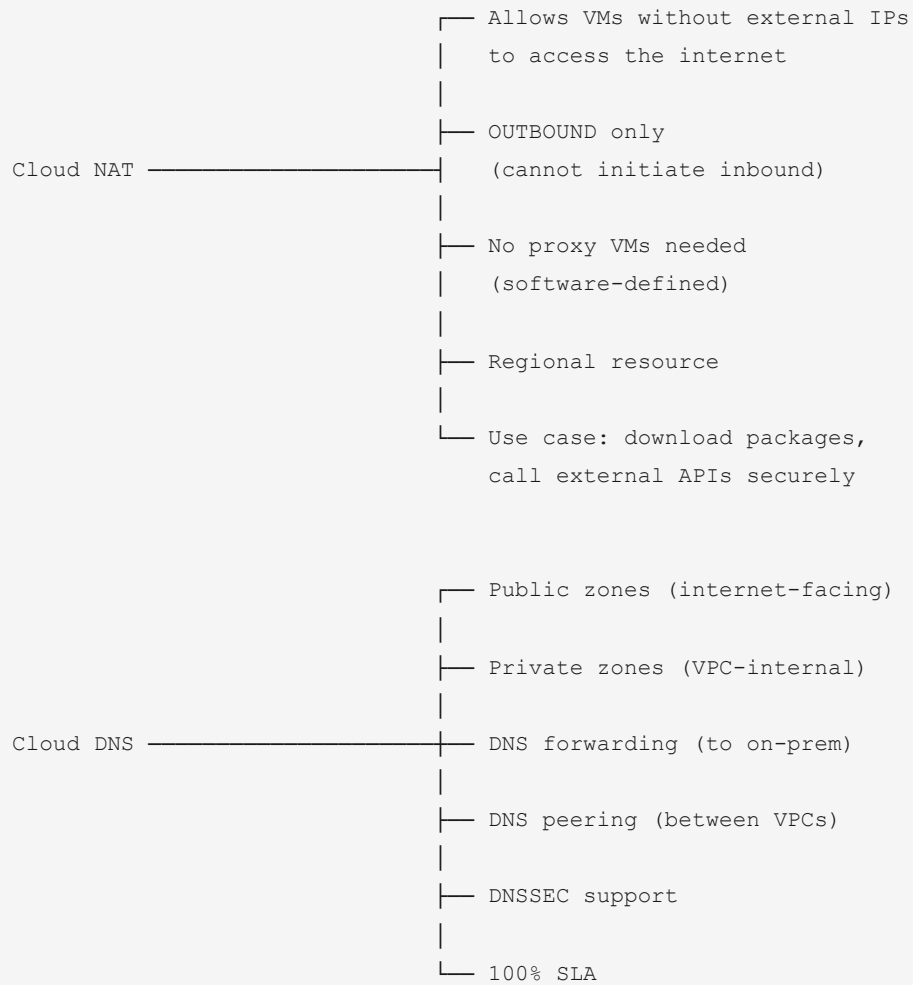
14. Connectivity to On-Premises

ON-PREMISES TO GCP CONNECTIVITY

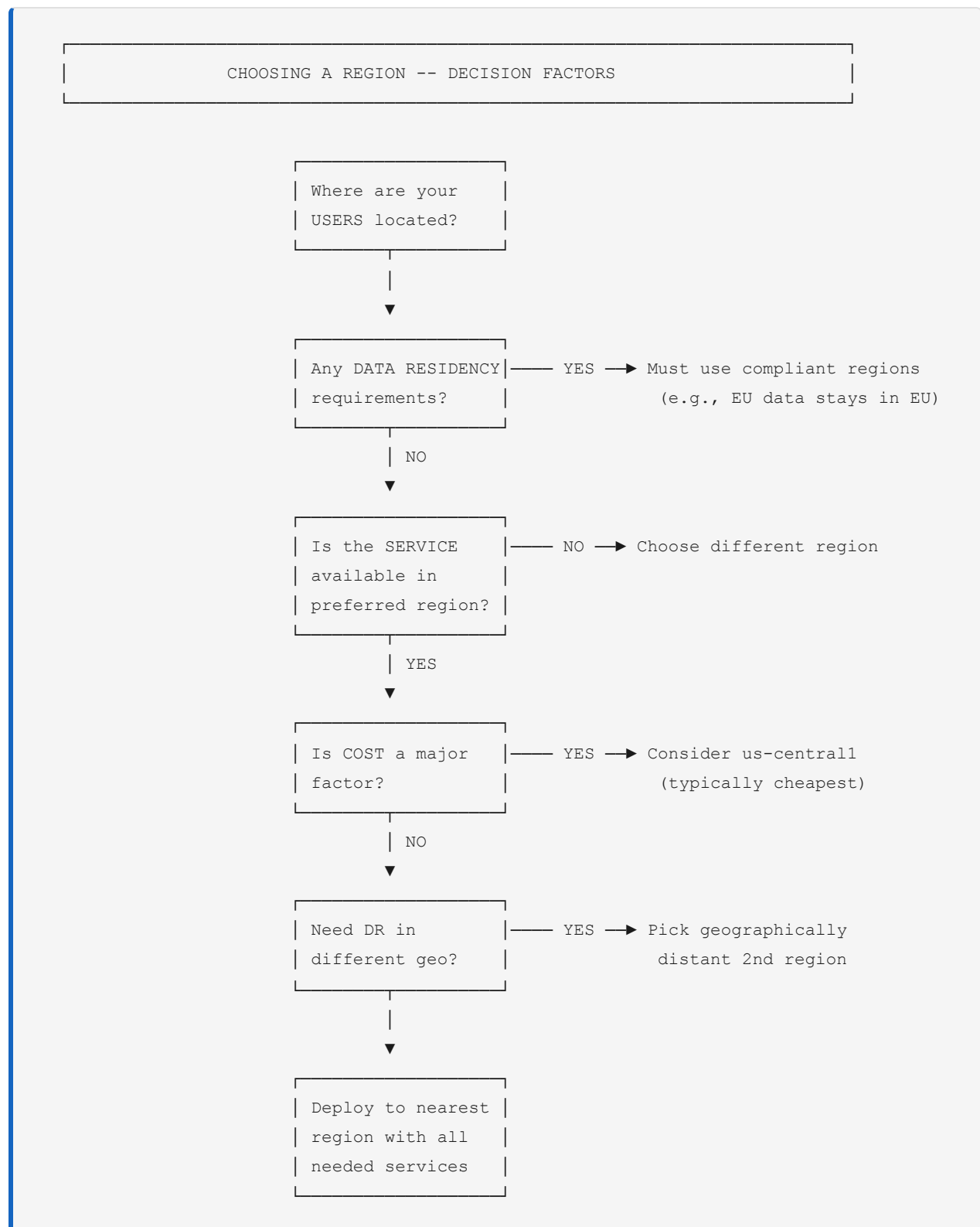
BANDWIDTH ← Low ————— High →

Classic VPN	HA VPN	Partner Interconnect	Dedicated Interconnect
1.4 Gbps per tunnel	1.4-3 Gbps per tunnel	50 Mbps - 50 Gbps	10-200 Gbps
99.9% SLA	99.99% SLA	99.9-99.99%	99.9-99.99%
Over public internet	Over public internet	Via service provider	Direct physical connection
Encrypted (IPsec)	Encrypted (IPsec)	Encrypted optional	NOT encrypted (add VPN on top)
Legacy Simple	Recommended Most common	Medium bandwidth	Enterprise Lowest latency

15. Cloud NAT and Cloud DNS Mind Maps



16. Region Selection Decision Process



17. Complete Section 2 Exam Cheat Sheet

SECTION 2: QUICK REFERENCE

COMPUTE:

Full control / GPU / Windows	Compute Engine
Kubernetes / orchestration / multi-cloud	GKE
Container, no K8s, stateless	Cloud Run
Event-driven, lightweight	Cloud Functions
Fault-tolerant batch + save money	Spot VMs
Exact CPU/RAM needs	Custom Machine Types

DATABASES:

SQL, <64TB, standard	Cloud SQL
PostgreSQL, HTAP, high perf	AlloyDB
SQL, global, unlimited scale	Spanner
Analytics, petabyte SQL	BigQuery
Mobile/web, real-time, documents	Firestore
Time-series, IoT, >1TB, low latency	Bigtable

STORAGE:

Hot data	Standard
<1x/month access	Nearline (30d min)
<1x/quarter access	Coldline (90d min)
<1x/year access	Archive (365d min)

ALL classes have MILLISECOND access latency!

NETWORKING:

Global web app LB	Ext. Application LB
Internal HTTP services	Int. Application LB
UDP / non-HTTP external	Ext. Passthrough Net LB
Low latency / production	Premium Tier
Cost-sensitive / dev	Standard Tier

VPC = Global | Subnet = Regional | VM = Zonal